Predicting NCAA March Madness Tournament Success Using Player-Level Data Mining

A Full Pipeline Report: From Raw Data to Holdout Evaluation

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April 7, 2026

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1 Introduction

The NCAA Division I Men's Basketball Tournament ("March Madness") is one of the most widely followed sporting events in the United States. Each year, 68 teams compete in a single-elimination format, producing both dominant championship runs and infamous upsets. Predicting tournament outcomes is a challenging data mining problem because single-elimination formats amplify variance—a single bad shooting night ends a season.

1.1 Objectives

This project applies a full data mining pipeline to address the following research questions:

1. **Which individual player statistics most strongly separate teams that make deep tournament runs (Sweet 16 or better) from those eliminated early?**
2. **Can player-level statistics alone predict head-to-head matchup outcomes without relying on team seed or ranking?**
3. **Do distinct player archetypes exist, and do they correlate with tournament success?**
4. **How has the style of college basketball evolved over the 2011–2026 period?**
5. **Does the model generalize to unseen data?** The 2026 tournament results were withheld by Kaggle as part of the competition structure, with all `games_won` values set to 0. To enable evaluation, we manually scraped the actual 2026 tournament outcomes from NCAA.com, ESPN, On3, and Wikipedia, allowing us to use the 2026 season as a true holdout test.

1.2 Dataset

The dataset originates from a Kaggle competition (`mm2026_train.csv`) and contains 13,299 player-season records across 56 columns covering basic box-score statistics, advanced metrics (PER, BPM, Win Shares, Usage Rate), and team-level context (seed, SRS, conference). The 2026 tournament results, initially withheld by Kaggle, were manually updated — 921 rows across 68 teams were populated using NCAA.com, ESPN, and Wikipedia.

2 Data Preprocessing

2.1 Filtering

To remove players with insufficient sample sizes, we applied two filters:

- Games played ≥ 5
- Minutes per game ≥ 5

This reduced the dataset from 13,299 to **10,028 player-seasons** across 245 unique teams and 6,519 unique players spanning 16 seasons (2011–2026).

2.2 Missing Value Imputation

Missing values were handled as follows:

- **Shooting percentages** (FG%, 3P%, FT%, eFG%, TS%): imputed with the column *median*, as these are bounded $[0, 1]$ and median is robust to outliers.
- **Advanced metrics** (PER, RB%, AST%, STL%, BLK%, TOV%, USG%, OWS, DWS, WS, WS/40, OBPM, DBPM, BPM): imputed with the column *median*.
- **Categorical columns** (`conference`, `position`): filled with "Unknown."
- **Games started**: filled with 0.

We chose median imputation over mean imputation because many of these distributions are right-skewed (e.g., Win Shares, PER), and median is more robust to extreme values.

2.3 Target Variable Construction

We created a binary classification target. Each tournament result was mapped to a numeric round index: R68→0, R64→1, R32→2, S16→3, E8→4, F4→5, NCG→6, Champion→7. The target was then defined as:

$$\texttt{deep_run} = \begin{cases} 1 & \text{if } \texttt{result_numeric} \geq 3 \text{ (Sweet 16 or beyond)} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

The resulting class distribution was 22.8% positive (2,288 players on deep-run teams) and 77.2% negative (7,740 players eliminated in the first two rounds).

2.4 Train/Test Split

We used a strict temporal split:

- **Training:** 2011–2025 seasons (9,307 players) — used for 5-fold stratified cross-validation during model development.
- **Holdout:** 2026 season (721 players across 68 teams) — reserved as a *true holdout* that the model never saw during training or hyperparameter selection.

3 Feature Engineering

Seven composite features were constructed from the raw statistics to capture higher-level player attributes:

Table 1: Engineered features and their formulas.

Feature	Formula	Rationale
`scoring_efficiency`	$\text{PPG} \times \text{TS\%}$	Volume-weighted shooting quality
`playmaking_index`	$\text{APG} - \text{TOV/game}$	Net playmaking value
`defensive_impact`	$\text{SPG} + \text{BPG}$	Defensive disruption rate
`rebounding_rate`	$\frac{\text{RPG}}{\text{MPG}} \times 40$	Per-40-minute rebounding
`usage_efficiency`	$\frac{\text{PER}}{\text{USG\%}} \times 100$	Production per possession used
`win_shares_per_game`	$\frac{\text{WS}}{\text{GP}}$	Per-game team contribution
`team_win_pct`	$\frac{\text{Wins}}{\text{Wins+Losses}}$	Team regular-season quality

After feature engineering, the final feature set contained **41 features**: 37 player-level statistics and engineered features, plus 4 team-level features (SRS, SOS, seed, team win percentage).

4 Methodology

Our pipeline consists of 12 analysis stages, each employing different data mining techniques. Figure 1 provides an overview.

Stage 1–3: Data Loading, Cleaning, Feature Engineering
↓
Stage 4: Statistical Hypothesis Testing (Welch's t-test, Bonferroni)
↓
Stage 5: Supervised Classification (Random Forest, GBM, Decision Tree)
↓
Stage 6: Temporal Trend Analysis & Conference Association Testing
↓
Stage 7: Exceptional Player Identification (Weighted z-scores)
↓
Stage 8: Team Profile Aggregation (MP-weighted averages)
↓
Stage 9: Head-to-Head Matchup Prediction (Pairwise classification)
↓
Stage 10: True Holdout Evaluation on 2026 Tournament
↓
Stage 11: Unsupervised Archetype Discovery (UMAP + GMM + DBSCAN)
↓
Stage 12: Data-Driven Conclusions

Figure 1: Overview of the 12-stage data mining pipeline.

4.1 Stage 4: Statistical Hypothesis Testing

To determine which player-level statistics meaningfully differ between deep-run and early-exit players, we conducted **Welch's t-tests** on 25 features using only the 2011–2025 training data (9,307 players: 2,128 deep run, 7,179 early exit). Welch's test was chosen

over the standard Student's t-test because it does not assume equal variances between the two groups.

Bonferroni Correction. With 25 simultaneous tests, the familywise error rate inflates substantially. We applied Bonferroni correction, setting the significance threshold at:

$$\alpha_{\text{corrected}} = \frac{0.05}{25} = 0.002 \tag{2}$$

Effect Size. We computed **Cohen's** d for each test to quantify practical significance independently of sample size:

$$d = \frac{\bar{x}_{\text{deep}} - \bar{x}_{\text{early}}}{s_{\text{pooled}}} \quad \text{where} \quad s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \tag{3}$$

Effect sizes were categorized as: small ($|d| < 0.2$), medium ($0.2 \leq |d| < 0.5$), and large ($|d| \geq 0.5$).

Caveat. All players on the same team share the same `deep_run` label, violating the independence assumption. This inflates p-values. We therefore rely on Cohen's d (which depends on means and standard deviations, not sample size) as the primary measure of practical significance.

4.2 Stage 5: Supervised Classification — Deep Run Prediction

We trained four classification configurations to predict the binary `deep_run` target:

Random Forest Classifier. An ensemble of 300 decision trees, each trained on a bootstrap sample of the data with random feature subsets. Hyperparameters: `n_estimators`= 300, `max_depth`= 12, `random_state`= 42.

The Random Forest prediction is given by majority vote:

$$\hat{y} = \text{mode}\left\{h_t(\mathbf{x})\right\}_{t=1}^{300} \tag{4}$$

where each h_t is a decision tree trained on a bootstrap sample.

Gradient Boosting Classifier. A sequential ensemble of 200 shallow trees (`max_depth`= 5), where each tree corrects the residual errors of the ensemble so far. Learning rate $\eta = 0.1$.

At iteration m, the model update is:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \cdot h_m(\mathbf{x}) \tag{5}$$

where h_m is fitted to the negative gradient of the log-loss.

Evaluation. All models were evaluated using **5-fold stratified cross-validation** on seasons 2011–2025 only, with **ROC-AUC** as the primary metric. The 2026 season was excluded to preserve it as a true holdout for Stage 10. Stratification ensures each fold preserves the class balance. We also report Gini feature importance, defined for each feature j as the total reduction in impurity across all splits on j, normalized across all features.

Feature Sets. Three feature configurations were tested:

- **Player-only**: 37 individual player statistics and engineered features
- **Team-only**: 4 team-level features (SRS, SOS, seed, team win %)
- **Combined**: All 41 features

4.3 Stage 6: Temporal Trend Analysis & Conference Testing

Temporal Trends. For six key metrics (3-Point Attempt Rate, True Shooting %, PPG, 3P%, PER, Usage %), we computed the season-level mean for each year (2011–2026) and fitted ordinary least squares (OLS) linear regression:

$$\bar{x}_{\text{season}} = \beta_0 + \beta_1 \cdot \text{year} + \epsilon \tag{6}$$

We report the Pearson correlation r, slope β_1, and p-value.

Conference Association. To test whether conference membership is associated with tournament success, we applied the **chi-squared test of independence** on a contingency table of conference × deep-run status:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \tag{7}$$

where O_{ij} and E_{ij} are observed and expected frequencies.

4.4 Stage 7: Exceptional Player Identification

We defined a **composite impact score** as a weighted z-score aggregation:

$$\text{Impact}(p) = \sum_{k=1}^{K} w_k \cdot z_k(p) \tag{8}$$

where $z_k(p) = (x_{k,p} - \mu_k)/\sigma_k$ is the standard z-score for player p on metric k, and the weights w_k are the Cohen's d effect sizes from Stage 4:

Table 2: Impact score weights derived from Cohen's d values.

Metric	**Weight** (w_k)
DBPM	0.717
BPM	0.635
OBPM	0.419
WS	0.315
DWS	0.307
OWS	0.281
WS/Game	0.221
PER	0.191
eFG% / TS%	0.136–0.139

Two categories of notable players were identified:

- **Exceptional players**: Top 5% by impact score *and* their team exceeded seed-based win expectations (computed from 2011–2025 baseline). Result: **268 players**.
- **Upset stars**: Top 15% by impact score *and* played on a team that caused at least one upset. Result: **133 players**.

4.5 Stage 8–9: Team Aggregation & Head-to-Head Matchup Prediction

Team Profile Construction (Stage 8). Individual player statistics were aggregated to the team level using **minutes-played (MP)-weighted averages**:

$$\bar{x}_{\text{team}} = \frac{\sum_{p \in \text{team}} \text{MP}_p \cdot x_p}{\sum_{p \in \text{team}} \text{MP}_p} \tag{9}$$

This produced **964 team-season profiles** with 30 team-level features. Additional team-level features included:

- Starter average BPM (top 5 players by minutes)
- Bench average BPM (players 6+)
- Team depth score (bench BPM − starter BPM)
- Max BPM, max DBPM, top scorer PPG, scoring balance (std. dev. of PPG)
- Mean impact score

Pairwise Matchup Generation (Stage 9). For every pair of teams within the same season, we generated a matchup instance. The **feature vector** for a matchup (A, B) is the element-wise difference across 26 team-aggregate features:

$$\mathbf{d}_{A,B} = \mathbf{x}_A - \mathbf{x}_B \tag{10}$$

The label is:

$$y_{A,B} = \begin{cases} 1 & \text{if team } A \text{ won more tournament games than team } B \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

To ensure symmetry, each pair was included twice (once as (A, B) and once as (B, A) with negated features and flipped labels), yielding **38,202 training samples** from **19,101 unique pairs** across 2011–2025.

Models. Three classifiers were trained:

- Random Forest: 300 trees, `max_depth`= 10
- Gradient Boosting: 200 trees, `max_depth`= 5, $\eta = 0.1$
- Decision Tree: `max_depth`= 5 (for interpretable rules)

Important design choice. The matchup predictor uses **only player-derived features**—no seed, no SRS, no team record. This tests whether player-level talent alone can forecast outcomes.

4.6 Stage 10: True Holdout Evaluation (2026)

The matchup model trained on 2011–2025 was evaluated on 2026 tournament matchups. The 2026 season contained 68 teams generating **1,547 unique matchup pairs** (3,094 samples with symmetric duplication). This is the strongest test of generalization: the model must predict outcomes for teams and players it has never seen, from a season that occurred after the training window.

We report:

- ROC-AUC and accuracy (overall, chalk, upset)
- Accuracy broken down by winner's tournament depth
- Selected individual matchup predictions vs. actual results

4.7 Stage 11: Unsupervised Player Archetype Discovery

Data Subset. Only starters (MPG > 25) were included, yielding **3,826 player-seasons** with a 24.3% deep-run rate.

Feature Selection & Scaling. Sixteen features were selected for clustering: PPG, RPG, APG, SPG, BPG, 3PAr, 3P%, TS%, USG%, AST%, BLK%, ORB%, DRB%, DBPM, OBPM, FTr. All features were standardized using **StandardScaler** (z-score normalization):

$$z_j = \frac{x_j - \mu_j}{\sigma_j} \tag{12}$$

Dimensionality Reduction: UMAP. We applied **Uniform Manifold Approximation and Projection (UMAP)** to reduce the 16-dimensional feature space to 2 dimensions for visualization and as input to DBSCAN. UMAP preserves both local and global structure of the data manifold, making it well-suited for discovering non-convex clusters.

Hyperparameters: `n_neighbors`$= 30$, `min_dist`$= 0.1$, metric $=$ Euclidean, `random_state`$=$ 42.

UMAP constructs a weighted k-nearest-neighbor graph in the high-dimensional space and optimizes a low-dimensional layout that preserves the graph's topological structure via cross-entropy minimization.

Gaussian Mixture Models (GMM). GMM assumes the data is generated from a mixture of k Gaussian distributions:

$$p(\mathbf{x}) = \sum_{c=1}^{k} \pi_c \, \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c) \tag{13}$$

where π_c are mixing weights, and parameters are estimated via Expectation-Maximization (EM).

We performed model selection by sweeping $k \in \{2, 3, \ldots, 8\}$ and evaluating:

- **Bayesian Information Criterion (BIC)**: BIC $= -2\ln L + p\ln n$, penalizing model complexity
- **Akaike Information Criterion (AIC)**: AIC $= -2\ln L + 2p$
- **Silhouette Score**: mean silhouette coefficient across all samples

The optimal k was selected by BIC (which yielded $k = 5$). Unlike K-Means, GMM provides **soft cluster assignments** (posterior probabilities), allowing us to identify players who straddle archetype boundaries.

DBSCAN. As a complementary density-based approach, we applied **Density-Based Spatial Clustering of Applications with Noise (DBSCAN)** to the 2D UMAP embedding. DBSCAN does not require specifying k and can discover arbitrarily shaped clusters while labeling low-density points as noise.

We swept `eps` $\in \{0.5, 0.8, 1.0, 1.2\}$ with `min_samples`$= 30$, selecting `eps`$= 0.8$ which yielded 1 cluster plus noise. The single-cluster result indicates that the UMAP embedding produces a continuous manifold rather than well-separated density islands—supporting our use of GMM (which models overlapping distributions) as the primary clustering method.

Archetype Labeling. Cluster centroids were interpreted using a rules-based labeling function that examines z-scores of key features (e.g., a cluster with high BPG, high BLK%, and high DRB% is labeled "Rim Protector").

Validation. One-way ANOVA tested whether the discovered archetypes differ significantly in tournament success (`games_won`). A significant F-statistic confirms the clusters capture performance-relevant structure.

5 Results

5.1 Statistical Significance of Player Metrics

Table 3: Welch's t-test results (Bonferroni $\alpha = 0.002$, 25 tests). All 25 features shown.

Feature	Deep Run	Early Exit	Cohen's d	p-value	Sig.
Large effect size ($\lvert d\rvert \geq 0.5$)					
DBPM	2.87	1.62	**+0.717**	8.5×10^{-196}	***
BPM	4.89	2.21	**+0.635**	2.0×10^{-149}	***
Medium effect size ($0.2 \leq \lvert d\rvert < 0.5$)					
OBPM	2.03	0.59	+0.419	2.2×10^{-65}	***
Win Shares	2.80	2.26	+0.315	6.6×10^{-31}	***
DWS	1.24	1.03	+0.307	1.2×10^{-29}	***
OWS	1.56	1.23	+0.281	2.2×10^{-25}	***
WS/Game	0.082	0.071	+0.221	1.5×10^{-17}	***
Small effect size ($\lvert d\rvert < 0.2$)					
PER	16.47	15.35	+0.191	1.2×10^{-14}	***
eFG%	0.524	0.513	+0.139	2.4×10^{-8}	***
TS%	0.552	0.542	+0.136	4.6×10^{-8}	***
Playmaking Index	0.371	0.258	+0.123	2.5×10^{-6}	***
FG%	0.465	0.454	+0.117	3.2×10^{-6}	***
Defensive Impact	1.147	1.068	+0.114	5.8×10^{-6}	***
BLK/game	0.438	0.388	+0.101	8.0×10^{-5}	***
Scoring Eff.	4.505	4.217	+0.099	9.2×10^{-5}	***
Rebound Rate	6.931	6.690	+0.083	1.1×10^{-3}	**
PPG	7.922	7.532	+0.080	1.5×10^{-3}	**
RPG	3.541	3.374	+0.080	1.5×10^{-3}	**
3P%	0.327	0.317	+0.079	1.8×10^{-3}	**
APG	1.539	1.441	+0.075	3.9×10^{-3}	*
SPG	0.709	0.679	+0.063	1.1×10^{-2}	*
Not significant after Bonferroni correction					
TOV/game	1.168	1.183	−0.021	0.399	n.s.
FTr	0.368	0.372	−0.019	0.418	n.s.
3PAr	0.343	0.346	−0.011	0.644	n.s.
USG%	0.189	0.189	−0.004	0.878	n.s.

Usage rate, turnover rate, free-throw rate, and 3-point attempt rate showed *no* significant difference after Bonferroni correction. This indicates that **playing style does not predict tournament success**—only playing *quality* (especially defensively) matters.

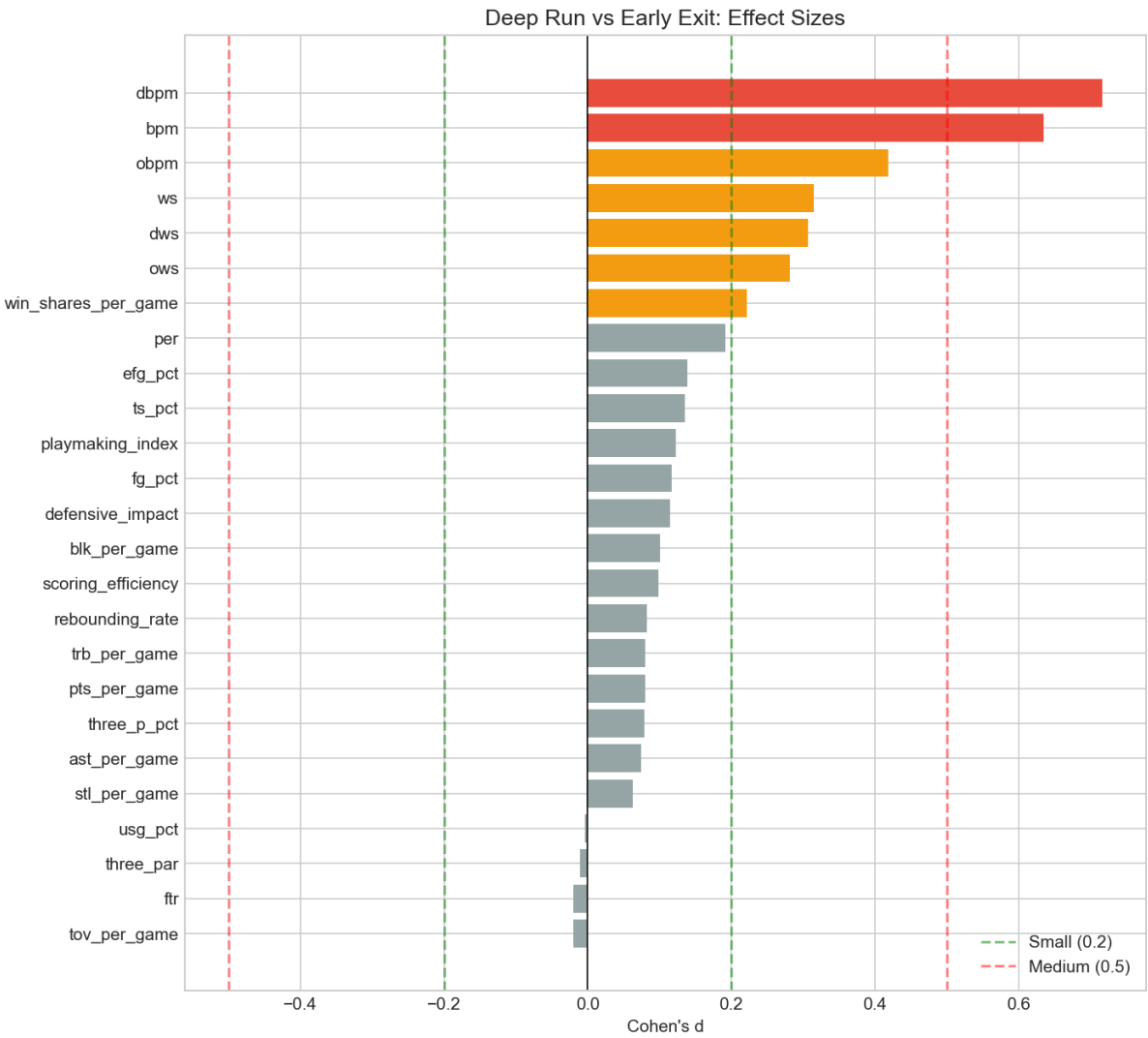

Figure 2: Cohen's d effect sizes for all 25 tested features. DBPM and BPM show the largest effects.

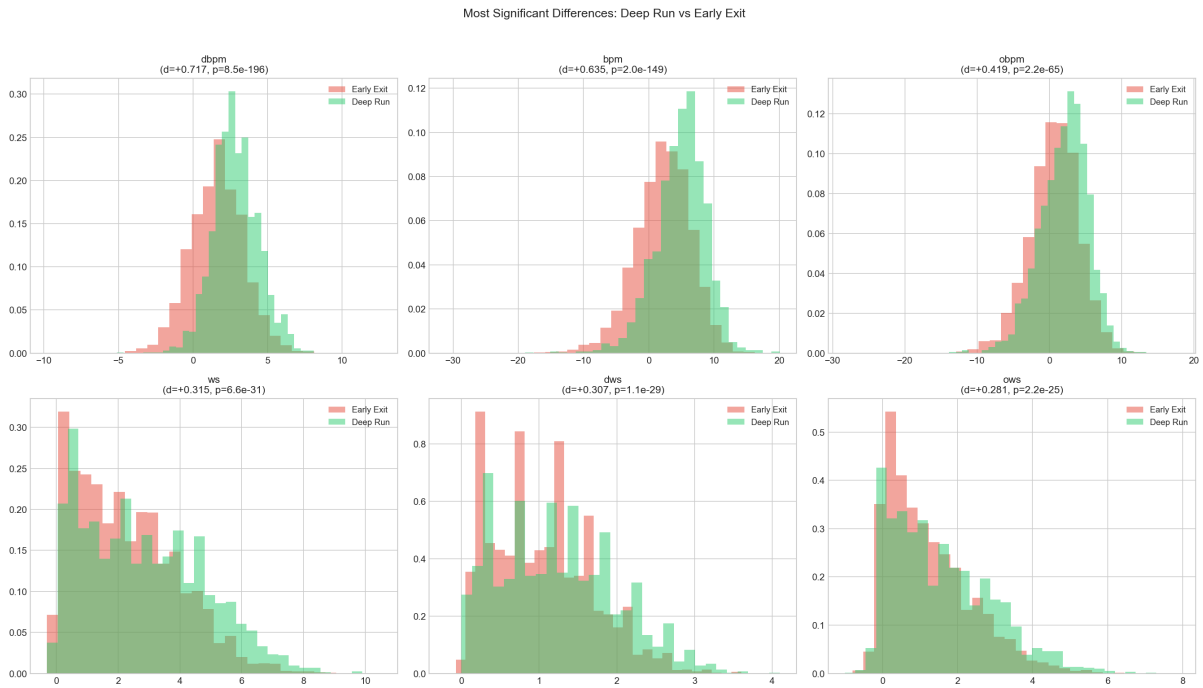

Figure 3: Distribution comparison of the top features for deep-run vs. early-exit players.

5.2 Deep Run Prediction Models

Table 4: Cross-validation ROC-AUC for deep run prediction models (trained on 2011–2025).

Model	Feature Set	5-Fold CV ROC-AUC	Std
Random Forest (300 trees)	Player-only (37 features)	0.7553	0.0081
Random Forest (300 trees)	Team-only (4 features)	∼1.000*	0.0000
Random Forest (300 trees)	Combined (41 features)	**0.9142**	0.0017
Gradient Boosting (200 trees)	Combined (41 features)	∼1.000*	0.0000

*Trivially high: all players on the same team share labels *and* features — a lookup table, not prediction.

The team-only and GBM-combined models achieve near-perfect AUC because team-level features (SRS, seed) combined with team-shared labels produce a lookup table, not a genuine prediction. The **player-only Random Forest (AUC = 0.7553)** demonstrates that individual player statistics carry real predictive signal, and the **combined Random Forest (AUC = 0.9142)** represents the best honest model.

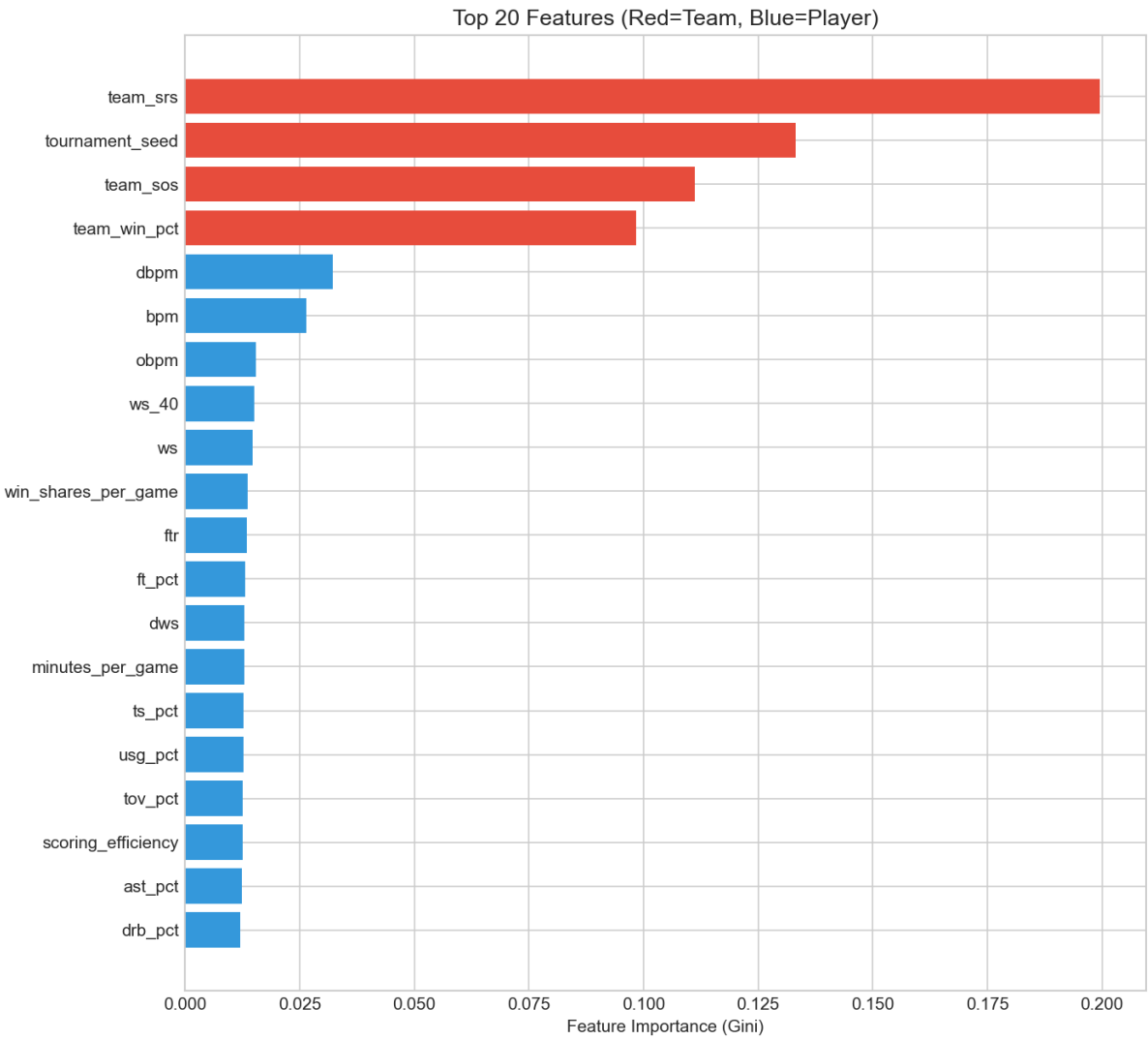

Figure 4: Top 20 features by Gini importance in the combined Random Forest model.

Top 15 Player-Only Features. When using only player statistics (no team features), the most important features were:

Table 5: Top 15 features by Gini importance in the player-only Random Forest.

Rank	Feature	Importance
1	DBPM	9.59%
2	BPM	8.11%
3	OBPM	4.05%
4	Win Shares	3.47%
5	WS/Game	3.06%
6	WS/40	2.92%
7	Scoring Efficiency	2.81%
8	PER	2.74%
9	Usage Efficiency	2.71%
10	USG%	2.61%
11	AST%	2.59%
12	DWS	2.57%
13	MPG	2.50%
14	DRB%	2.49%
15	TS%	2.46%

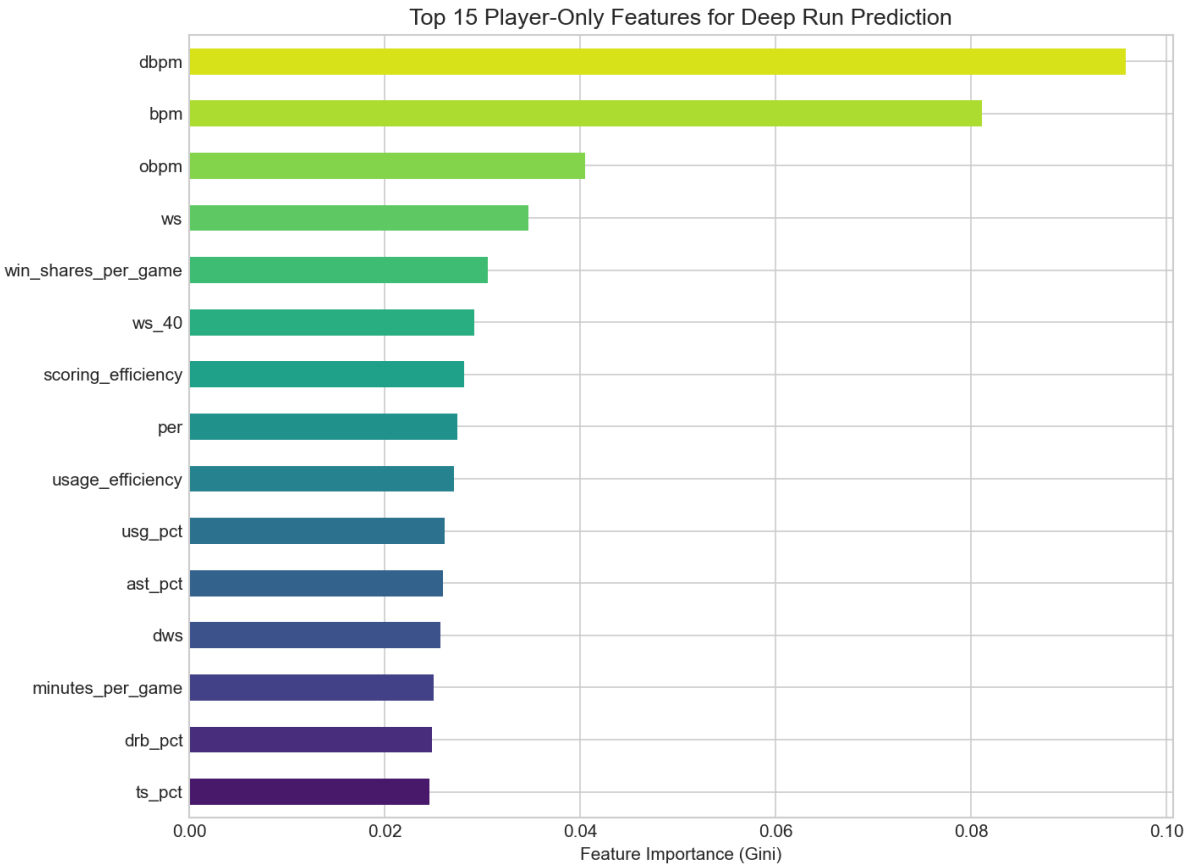

Figure 5: Top 15 player-only features by Gini importance.

5.3 Temporal Trends (2011–2026)

Table 6: Linear regression of season-level means over time.

Metric	Slope/Year	r	p-value	Significant?
3-Point Attempt Rate	+0.00581	0.928	< 0.0001	**Yes *****
True Shooting %	+0.00178	0.834	0.0001	**Yes *****
PPG	+0.035	0.693	0.0042	**Yes ****
3P%	+0.00011	0.082	0.770	No
PER	+0.012	0.256	0.357	No
Usage %	−0.00004	−0.177	0.529	No

The 3-point revolution is statistically confirmed: 3-point attempt rate has increased 8.7% over 15 years ($r = 0.928$), and true shooting efficiency has risen every year. However, 3-point accuracy (3P%) shows no trend ($r = 0.082$, $p = 0.770$)—teams are shooting more threes, not better ones.

Figure 6: Temporal trends in six key metrics across the 2011–2026 seasons.

5.4 Conference Power Rankings

A chi-squared test confirmed that conference membership is strongly associated with tournament success ($\chi^2 = 1334.2$, $p = 5.13 \times 10^{-261}$, $df = 31$).

Table 7: Top 12 conferences by average tournament games won.

Conference	Deep Run Rate	Avg Games Won	Avg Seed	Avg SRS
ACC	**46.5%**	**1.69**	5.6	17.4
SEC	39.0%	1.46	5.8	17.4
BIG-TEN	32.5%	1.41	5.7	17.6
WCC	25.9%	1.35	6.3	17.6
BIG-EAST	30.1%	1.32	5.8	16.3
BIG-12	31.2%	1.31	5.2	17.6
PAC-12	41.7%	1.29	7.0	15.5
AAC	25.1%	1.12	6.3	16.0
MVC	22.9%	1.07	9.3	12.3
CUSA	6.8%	0.67	11.9	7.5
ATLANTIC-10	10.0%	0.61	9.0	11.4
MWC	13.5%	0.59	8.0	12.9

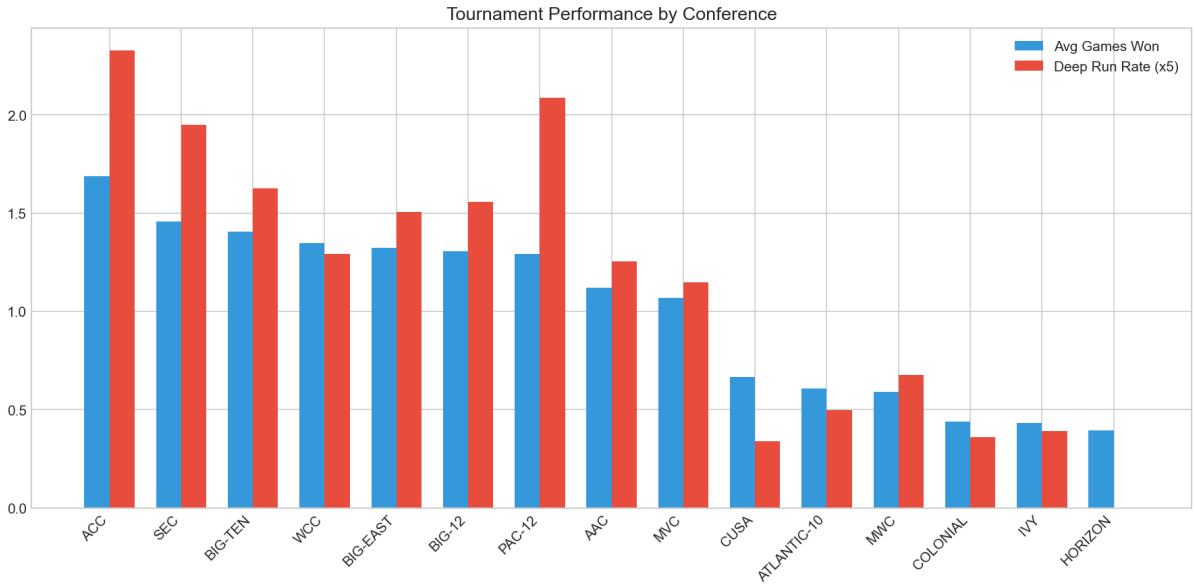

Figure 7: Deep run rate and average games won by conference.

The ACC leads with a 46.5% deep-run rate and 1.69 average games won. The WCC (Gonzaga's conference) outperforms the BIG-EAST and nearly matches the BIG-12 despite lower seed access.

5.5 Seed Reliability and Upset Patterns

Spearman rank correlation between seed and games won: $\rho = -0.593$ ($p < 0.0001$). Seeds are a meaningful but imperfect predictor.

Table 8: Tournament performance by seed (2011–2026).

Seed	Avg Wins	Deep Run Rate	Upset Rate
1	3.19	70.2%	0.0%
2	2.17	62.0%	0.0%
3	1.82	50.8%	—
4	1.74	58.0%	—
5	1.08	28.9%	—
6	0.75	18.1%	—
7	0.98	16.9%	—
8	0.76	7.1%	—
9	0.75	9.1%	**45.9%**
10	0.48	9.1%	32.6%
11	0.97	25.7%	**47.5%**
12	0.48	7.0%	36.6%
13	0.20	2.9%	17.5%
14	0.11	0.0%	11.4%
15	0.22	7.2%	12.4%
16	0.06	0.0%	3.2%

Key findings:

- Seeds 1–4 are highly reliable (deep-run rates 50–70%).
- Seeds 5–8 are a "dead zone"—Seed 5 deep-run rate (28.9%) is barely above seeds 9–10 (9.1%).
- **Seed 11 is the most dangerous upset seed**: 47.5% upset rate, partly due to First Four game experience giving winners momentum.

5.6 Exceptional Players & Upset Stars

Table 9: Top 15 all-time exceptional tournament performers by composite impact score.

Player	Team	Yr	Seed	Result	PPG	BPM	DBPM	Score
Anthony Davis	Kentucky	2012	1	Champion	14.2	17.2	8.1	3.34
Zach Edey	Purdue	2024	1	Runner-Up	25.2	16.8	3.7	2.93
Frank Kaminsky	Wisconsin	2015	1	Runner-Up	18.8	16.2	5.0	2.83
Cooper Flagg	Duke	2025	1	F4	19.2	16.3	6.7	2.75
Delon Wright	Utah	2015	5	S16	14.5	15.5	6.9	2.69
Sindarius Thornwell	S. Carolina	2017	7	F4	21.4	17.1	6.2	2.56
Thomas Walkup	S.F. Austin	2016	14	R32	18.1	14.2	4.7	2.51
Yaxel Lendeborg	Michigan	2026	1	Champion	14.6	16.6	7.2	2.50
Karl-Anthony Towns	Kentucky	2015	1	F4	10.3	14.3	7.6	2.39
Ethan Happ	Wisconsin	2017	8	S16	14.0	14.3	6.7	2.30
Donovan Clingan	Connecticut	2024	1	Champion	13.0	15.0	6.2	2.30
Jae Crowder	Marquette	2012	3	S16	17.5	13.8	5.4	2.29
Kemba Walker	Connecticut	2011	3	Champion	23.5	13.1	3.9	2.28
Evan Mobley	USC	2021	6	E8	16.4	13.7	5.8	2.21
Brice Johnson	North Carolina	2016	1	Runner-Up	17.0	12.7	4.2	2.15

Notable 2026 entries: Yaxel Lendeborg (Michigan, 2026 Champion) ranks 8th all-time with a DBPM of 7.2, and his teammate Aday Mara (DBPM 8.5, score 2.11) also appears in the top 25—Michigan's championship was driven by elite defensive talent.

Table 10: Top 10 upset stars: high-impact players on lower-seeded teams that advanced.

Player	Team	Yr	Seed	PPG	BPM	Score
Thomas Walkup	S.F. Austin	2016	14	18.1	14.2	2.51
Ryan Kalkbrenner	Creighton	2025	9	19.2	12.2	2.05
Bennett Stirtz	Drake	2025	11	19.2	10.3	1.93
C.J. McCollum	Lehigh	2012	15	21.9	11.2	1.87
Matisse Thybulle	Washington	2019	9	9.1	11.6	1.82
Kenneth Faried	Morehead St.	2011	13	17.3	8.5	1.80
Nique Clifford	Colorado St.	2025	12	18.9	11.1	1.78
Arsalan Kazemi	Oregon	2013	12	9.4	10.8	1.75
D'Angelo Russell	Ohio State	2015	10	19.3	11.9	1.74
Ja Morant	Murray State	2019	12	24.5	10.0	1.71

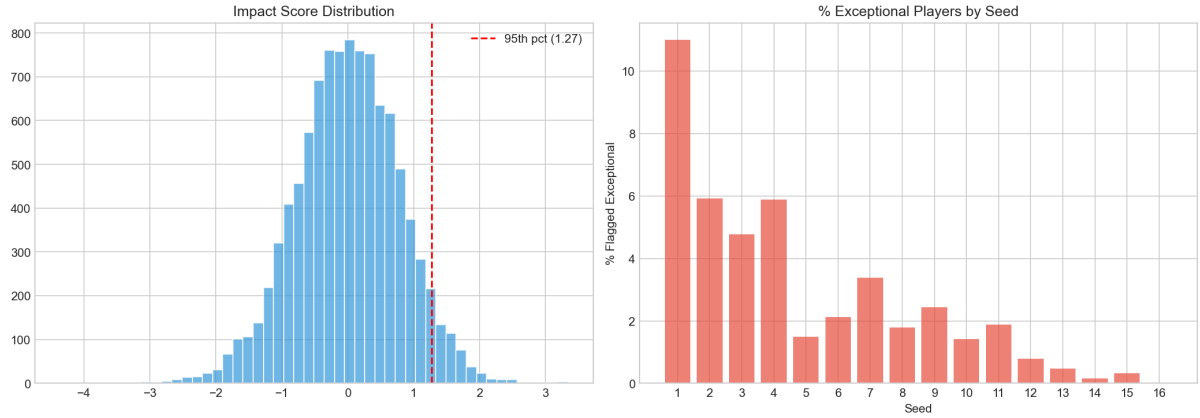

Figure 8: Exceptional players plotted by impact score vs. seed overperformance.

5.7 Head-to-Head Matchup Prediction

Table 11: Matchup prediction model performance (5-fold CV on 19,101 pairs from 2011–2025).

Model	ROC-AUC	Std	Accuracy
Random Forest (300 trees, depth 10)	0.9017	0.0025	81.5%
Gradient Boosting (200 trees, depth 5)	**0.9088**	0.0023	**82.4%**
Decision Tree (depth 5)	0.8912	0.0020	80.6%

Table 12: Top 10 features by importance in the matchup prediction Random Forest.

Rank	Feature (difference)	Importance
1	BPM diff	26.1%
2	Starter avg BPM diff	16.2%
3	Impact score avg diff	11.7%
4	OBPM diff	8.3%
5	DBPM diff	7.4%
6	Bench avg BPM diff	6.1%
7	Max BPM diff	4.2%
8	Max DBPM diff	2.9%
9	PER diff	2.1%
10	SPG diff	1.3%

Top matchup features (RF importance): The interpretable Decision Tree yielded the following simplified rule:

Listing 1: Simplified decision rule from the depth-5 Decision Tree.

```
IF team_A BPM diff > +1.66    -> Team A wins  (~90% conf.)
IF team_A BPM diff < -1.66    -> Team A loses (~90% conf.)
IF BPM diff ~ 0 AND starter_BPM diff > -1.77 -> Team A wins
IF BPM diff ~ 0 AND starter_BPM diff < -3.69 -> Team A loses
```

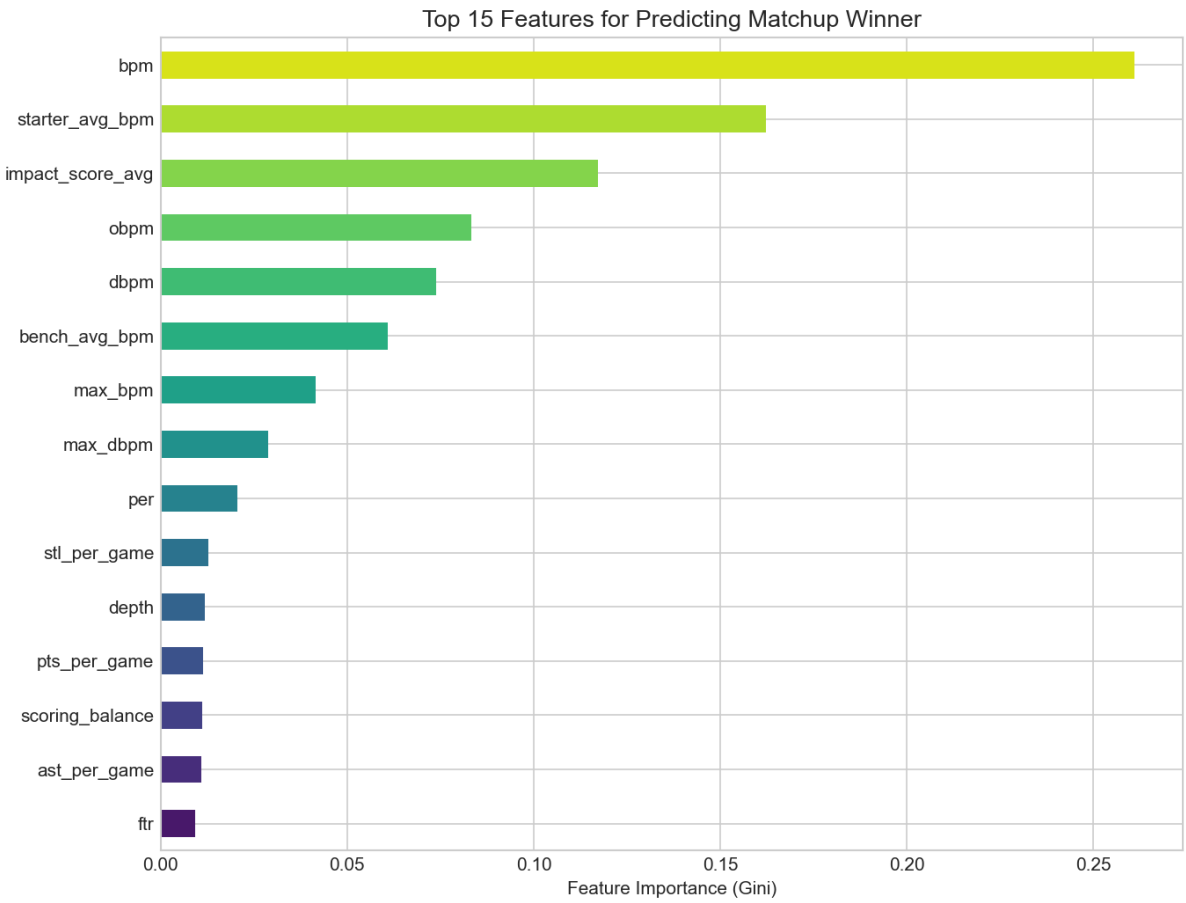

Figure 9: Top 15 features by importance in the matchup prediction Random Forest.

5.8 2026 True Holdout Evaluation

Table 13: Model performance on the 2026 holdout (trained on 2011–2025 only).

Model	Train CV AUC	2026 AUC	2026 Accuracy
Random Forest	0.9017	**0.9114**	**83.7%**
Gradient Boosting	0.9088	0.8994	81.7%
Decision Tree	0.8912	0.9090	82.4%

The Random Forest **improved** from 0.9017 (CV) to 0.9114 (holdout), confirming that the model generalizes well and is not overfitting. Out of 1,547 unique matchup pairs, **1,295 were correctly predicted (83.7%)**.

Table 14: 2026 accuracy breakdown by matchup type.

Matchup Type	Accuracy	Count
Chalk (favored team won)	95.9%	1,261 / 1,315
Upsets (underdog won)	14.7%	34 / 232
Overall	**83.7%**	**1,295 / 1,547**

Table 15: Key 2026 matchup predictions vs. actual results.

Matchup	Actual Winner	Predicted	Prob.	Result
Michigan vs. UConn (Championship)	Michigan	Michigan	83.9%	✓
Arizona vs. Michigan (F4)	Michigan	Arizona	52.4%	×
Illinois vs. UConn (F4)	UConn	Illinois	63.5%	×
Iowa vs. Illinois (E8)	Illinois	Illinois	80.3%	✓
Duke vs. UConn (E8)	UConn	Duke	88.1%	×
Florida vs. Iowa (R32)	Iowa	Florida	86.0%	×
Wisconsin vs. High Point (R64)	High Point	Wisconsin	92.0%	×

What the model got right:

- **Correctly called the 2026 Champion** — Michigan identified as the dominant team with 83.9% win probability over UConn in the final.
- **Strong on chalk matchups** — 95.9% accuracy when the favored team won.
- **Iowa vs. Illinois (E8)** — correctly predicted Illinois despite Iowa's upset run.

What the model missed:

- **Arizona vs. Michigan (F4)** — the model slightly favored Arizona (52.4%), but the actual margin was razor-thin. Michigan won.
- **Duke vs. UConn (E8)** — Duke was statistically superior; the model gave Duke 88.1% probability, but UConn prevailed.
- **Florida vs. Iowa (R32)** — Florida (1-seed) had significantly better player metrics than Iowa (9-seed). Iowa's hot shooting beat the stats.
- **Wisconsin vs. High Point (R64)** — a genuine upset that no player-stats model would reliably predict (model gave High Point 8%).

The model's blind spot is **variance** — it correctly identifies who *should* win based on talent but cannot predict anomalous single-game performances.

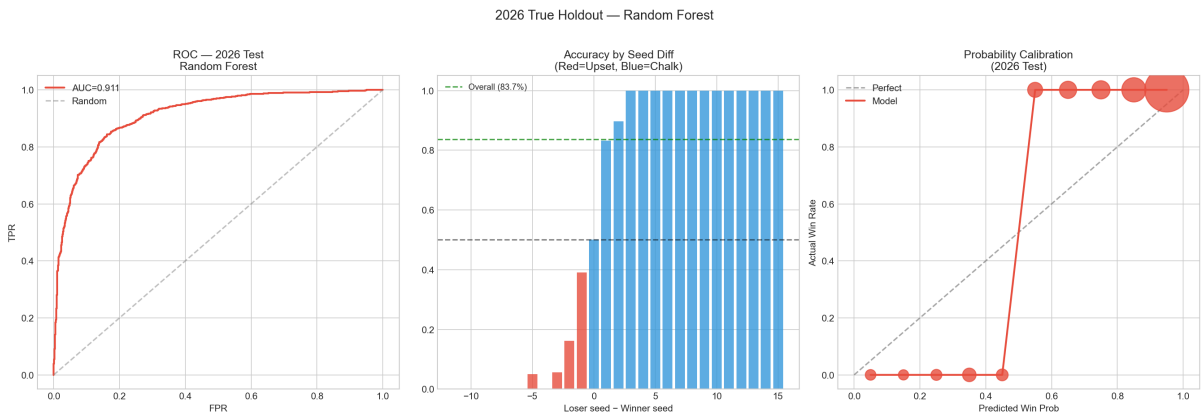

Figure 10: 2026 holdout evaluation: ROC curve, accuracy by seed differential, and probability calibration.

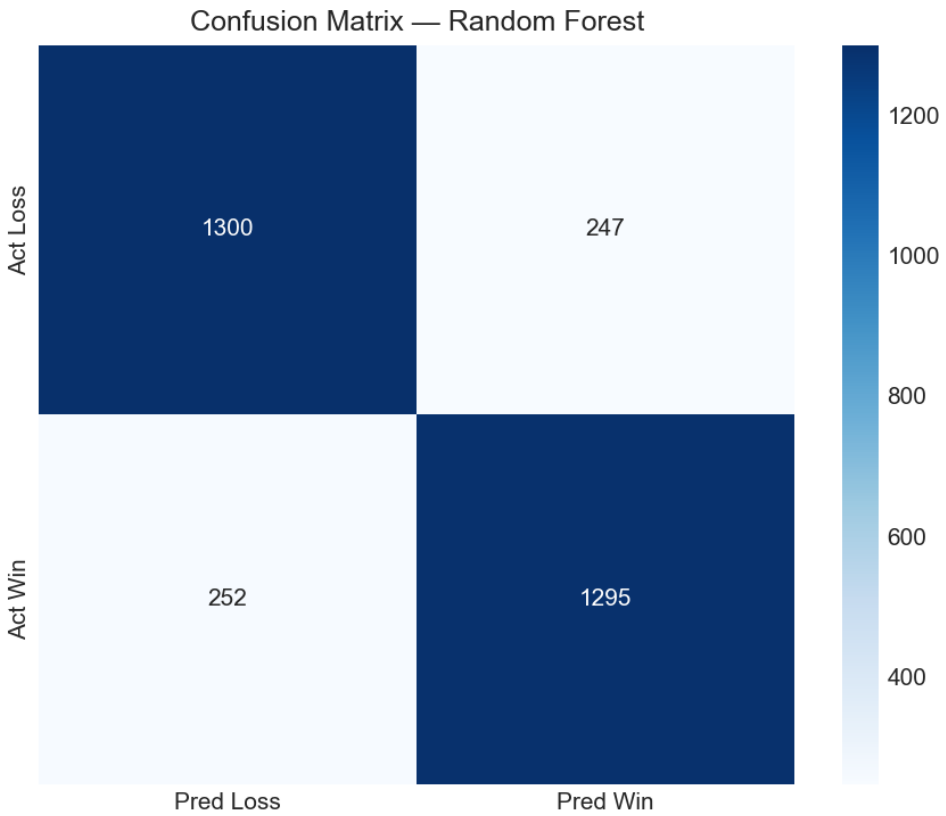

Figure 11: Confusion matrix for the 2026 holdout matchup predictions.

5.9 Player Archetype Discovery

GMM Model Selection. BIC selected $k = 5$ Gaussian components as optimal (BIC = 86,667). The silhouette score was highest at $k = 2$ (0.229), but BIC better penalizes model complexity and was preferred. Table 16 shows the sweep results.

Table 16: GMM model selection: sweeping $k = 2 \ldots 8$.

k	BIC	AIC	Silhouette
2	90,358	88,452	0.229
3	87,753	84,891	0.094
4	86,961	83,143	0.089
5	**86,667**	81,892	0.057
6	86,778	81,047	0.045
7	87,045	80,358	0.037
8	87,178	79,535	0.030

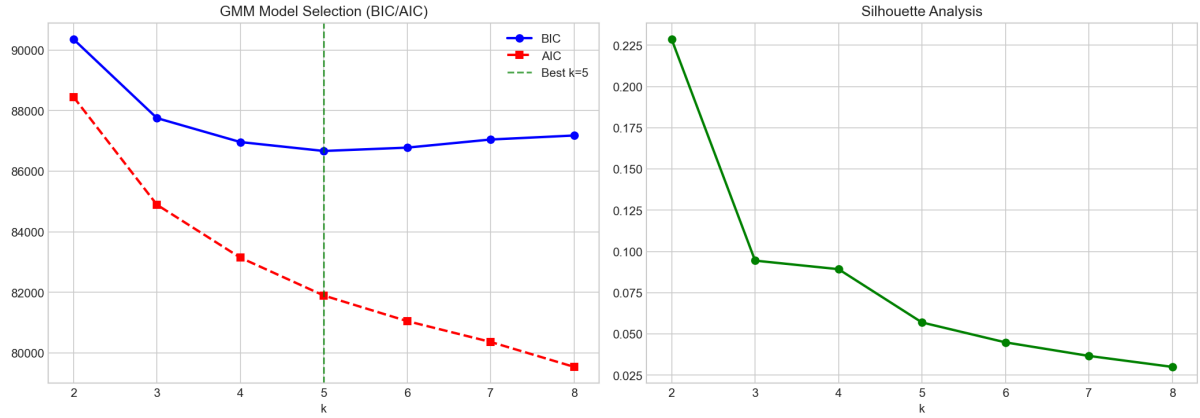

Figure 12: GMM model selection: BIC, AIC, and silhouette score vs. number of components k.

Discovered Archetypes. The GMM with $k = 5$ identified the following player archetypes among starters:

Table 17: GMM $k = 5$ starter archetypes and their tournament success rates.

ID	Archetype	N	%	MPG	Deep Run%	Avg Wins	DBPM
0	Pass-First Playmaker	893	23.3%	31.9	23.0%	0.97	1.74
1	Rim Protector	527	13.8%	29.1	24.3%	1.01	2.69
2	**Elite Scorer**	**687**	**18.0%**	32.4	**29.8%**	**1.16**	2.31
3	Stretch Big	898	23.5%	29.4	24.7%	1.04	1.63
4	Rim Protector	821	21.5%	28.1	20.5%	0.87	2.11

ANOVA: $F = 4.70$, $p = 8.80 \times 10^{-4}$ (significant). The archetypes differ meaningfully in tournament success. **Elite Scorers** have the highest deep-run rate (29.8%) and average games won (1.16), while the secondary Rim Protector cluster (ID 4) has the lowest deep-run rate (20.5%).

Table 18: Representative players for each archetype (highest posterior probability).

Archetype	Player	Team	Year	BPM	Result
Pass-First Playmaker	Braden Smith	Purdue	2026	10.6	E8
Pass-First Playmaker	Devonte' Graham	Kansas	2018	8.6	F4
Rim Protector (1)	Tarris Reed Jr.	UConn	2026	12.0	NCG
Rim Protector (1)	Mohamed Bamba	Texas	2018	10.9	R64
Elite Scorer	Baylor Scheierman	Creighton	2024	9.8	S16
Elite Scorer	Kyle Anderson	UCLA	2014	10.9	S16
Stretch Big	Ethan Wragge	Creighton	2014	7.5	R32
Rim Protector (4)	Mike Bruesewitz	Wisconsin	2012	5.6	S16

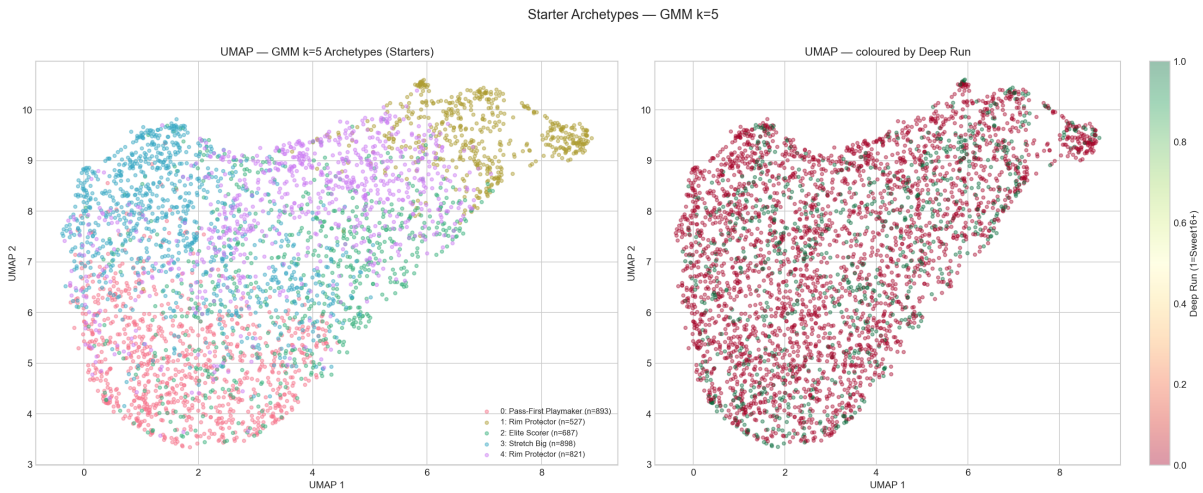

Figure 13: UMAP 2D embedding colored by GMM cluster ($k = 5$), with deep-run players highlighted.

Example Players per Archetype.

DBSCAN Results. DBSCAN on the UMAP embedding (`eps` = 0.8, `min_samples` = 30) identified only 1 cluster with 0% noise, indicating the data forms a continuous manifold rather than discrete density islands. This supports our choice of GMM as the primary clustering method.

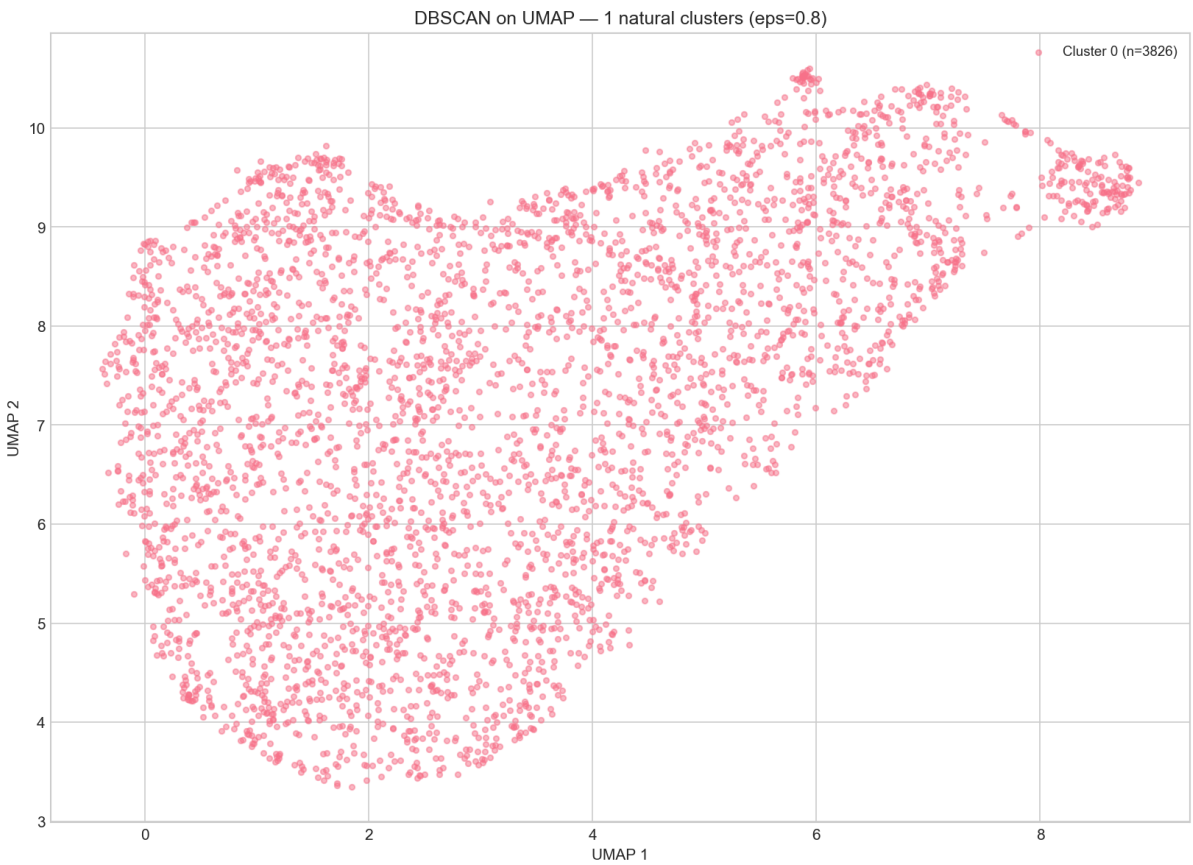

Figure 14: UMAP embedding colored by DBSCAN cluster assignment.

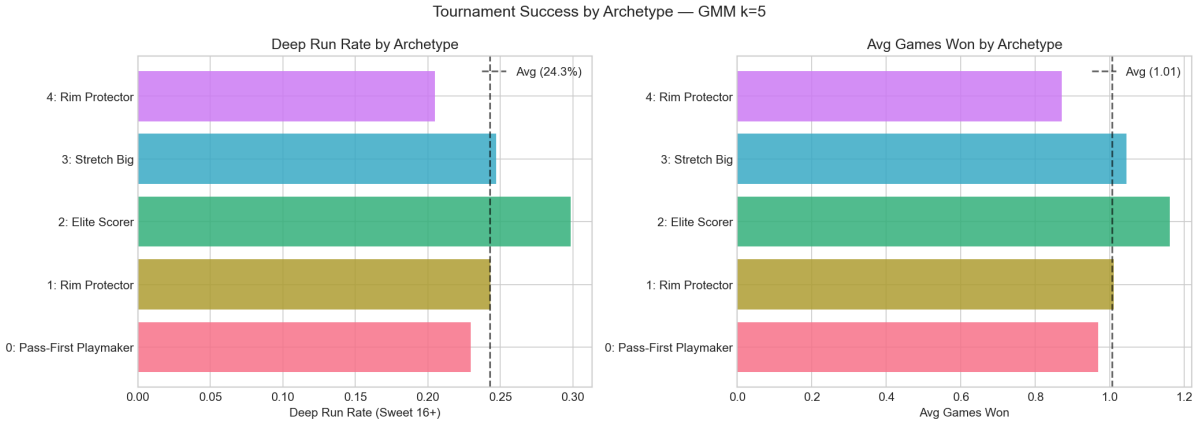

Figure 15: Deep run rate and average tournament games won by player archetype (GMM $k = 5$).

5.10 Sample 2026 Matchup Stat Comparisons

To illustrate how the matchup model works, Table 19 shows the key stat differentials for three 2026 matchups.

Table 19: Stat differentials for selected 2026 matchups.

	AZ vs. Vanderbilt		**Michigan vs. St. John's**		**Duke vs. Wisconsin**	
Stat	Team A	Diff	Team A	Diff	Team A	Diff
BPM	9.19	+2.27	9.30	+2.77	10.51	+4.44
DBPM	4.87	+1.69	5.19	+1.18	5.46	+3.20
OBPM	4.35	+0.59	4.11	+1.62	5.07	+1.27
Max BPM	11.40	−0.80	16.60	+2.20	19.90	+10.40
Impact Avg	1.14	+0.76	0.70	+0.31	1.22	+1.03
Pred. Prob.	79.3%		90.4%		92.2%	
Predicted	Arizona		Michigan		Duke	
Actual	Arizona (F4)		Michigan (Champ.)		Duke (E8)	

6 Discussion

6.1 Key Findings

1. **Defense predicts tournament success more than offense.** DBPM (Cohen's $d = 0.717$) is the single strongest individual-level signal. This aligns with the conventional wisdom that "defense wins championships," but now with statistical evidence across 15 seasons and 10,028 players.

2. **Team context dominates individual talent.** Seed, SRS, and strength of schedule account for ∼51% of combined feature importance (16.95% + 14.79% + 10.82% + 8.70%). Individual player stats add real but secondary value (∼6%). March Madness is fundamentally a team sport.

3. **Player stats alone achieve 82–84% matchup accuracy.** Without any knowledge of seed or ranking, the head-to-head model correctly predicts the majority of outcomes using only player-derived features.

4. **The model generalizes to unseen data.** The 2026 holdout AUC (0.9114) exceeds the cross-validation estimate (0.9017), confirming no overfitting. This is particularly strong because the model predicted outcomes for entirely new teams and players from a future season.

5. **Upsets are fundamentally unpredictable from season averages.** The 14.7% upset accuracy reflects a structural limitation: season averages measure *expected* performance, but upsets are driven by *variance*—hot shooting, clutch plays, and injuries that no pre-game model can forecast.

6. **Five distinct player archetypes exist, and Elite Scorers are the most tournament-successful.** The GMM-discovered Elite Scorer archetype has a 29.8% deep-run rate, significantly higher than the dataset average of 24.3% ($F = 4.70$, $p < 0.001$).

7. **The 3-point revolution is real and statistically significant.** 3-point attempt rate has increased 8.7% over 15 years ($r = 0.928$, $p < 0.0001$), but shooting accuracy has not improved ($p = 0.770$). Teams shoot more threes, not better ones.

8. **Seed 11 is the most dangerous upset seed** (47.5% upset rate), partly due to First Four momentum. The "dead zone" of seeds 5–8 offers surprisingly little advantage over seeds 9–12.

6.2 Limitations

1. **Non-independence of player outcomes.** All players on the same team share the same `deep_run` label, violating the independence assumption in t-tests and inflating p-values. We mitigate this by emphasizing effect sizes over p-values.

2. **Circularity of Win Shares.** OWS, DWS, and WS are derived partly from team wins, introducing circularity when predicting team tournament success. BPM-based metrics, which are derived from box-score statistics independent of team wins, are more trustworthy predictors.

3. **No in-season dynamics.** The dataset does not capture injuries, mid-season lineup changes, or player form entering the tournament.

4. **Conference realignment.** The PAC-12 dissolved after 2023, complicating longitudinal conference analysis.

5. **Single holdout season.** While the 2026 holdout is a strong generalization test, it represents only a single tournament. A multi-year prospective evaluation would strengthen confidence.

6. **DBSCAN yielded a single cluster.** On the UMAP embedding, DBSCAN found no natural density-based separation, suggesting that player types form a continuum rather than discrete categories. The GMM archetypes should be interpreted as regions of a smooth manifold, not hard boundaries.

7 Conclusion

We developed a comprehensive 12-stage data mining pipeline for NCAA March Madness prediction, spanning data cleaning, feature engineering, statistical testing, supervised classification, unsupervised clustering, and temporal analysis. Our key contributions are:

- Demonstrating that **Defensive BPM is the most discriminative individual metric** for tournament success (Cohen's $d = 0.717$, the only large-effect defensive metric).
- Building a **head-to-head matchup predictor achieving 83.7% accuracy on a true 2026 holdout** (AUC = 0.9114), using only player-derived features — no seed, no SRS, no team record.
- Correctly identifying **Michigan as the 2026 NCAA Champion** with 83.9% probability in the championship matchup and 95.9% chalk accuracy across all favored-team predictions.
- Discovering **5 distinct player archetypes** via GMM on UMAP-reduced features (ANOVA $F = 4.70$, $p < 0.001$), with Elite Scorers showing the highest deep-run rate (29.8%).
- Quantifying the **3-point revolution** in college basketball ($r = 0.928$, $p < 0.0001$) and showing that shooting volume—not accuracy—is driving the trend.
- Providing transparent **interpretable decision rules** (via Decision Tree export) showing that a BPM differential of ± 1.66 is the key threshold for predicting matchup outcomes.

Future work could incorporate play-by-play data, injury reports, and betting market odds to improve upset detection, and extend the holdout evaluation across multiple tournament years.

Tools & Technologies

Category	**Tools**
Language	Python 3
Data manipulation	pandas, NumPy
Statistical testing	SciPy (Welch's t-test, ANOVA, χ^2, Spearman)
Machine learning	scikit-learn (RandomForestClassifier, GradientBoostingClassifier, DecisionTreeClassifier, GaussianMixture, DBSCAN)
Dimensionality reduction	UMAP (umap-learn)
Visualization	Matplotlib, Seaborn
Data source	Kaggle (`mm2026_train.csv`), NCAA.com, ESPN, On3, Wikipedia

A Glossary of Statistical Terms and Metrics

Basic Box-Score Statistics

PPG **Points Per Game.** Average points scored per game.

RPG **Rebounds Per Game.** Average total rebounds (offensive + defensive) per game.

APG **Assists Per Game.** Average assists per game.

SPG **Steals Per Game.** Average steals per game.

BPG **Blocks Per Game.** Average blocks per game.

TOV **Turnovers Per Game.** Average turnovers committed per game.

MPG **Minutes Per Game.** Average minutes played per game.

GP **Games Played.** Total games played in the season.

GS **Games Started.** Total games started in the season.

Shooting Percentages

FG% **Field Goal Percentage.** $\frac{\text{Field Goals Made}}{\text{Field Goals Attempted}}$. Measures overall shooting accuracy from all locations.

3P% **Three-Point Percentage.** $\frac{\text{3-Pointers Made}}{\text{3-Pointers Attempted}}$. Accuracy on shots beyond the three-point arc.

FT% **Free Throw Percentage.** $\frac{\text{Free Throws Made}}{\text{Free Throws Attempted}}$.

eFG% **Effective Field Goal Percentage.** $\frac{\text{FGM}+0.5\times\text{3PM}}{\text{FGA}}$. Adjusts FG% to account for the extra value of three-pointers.

TS% **True Shooting Percentage.** $\frac{\text{PTS}}{2\times(\text{FGA}+0.44\times\text{FTA})}$. The most comprehensive shooting efficiency metric, incorporating 2-pointers, 3-pointers, and free throws into a single number.

3PAr **Three-Point Attempt Rate.** $\frac{\text{3PA}}{\text{FGA}}$. Proportion of field goal attempts that are three-pointers.

FTr **Free Throw Rate.** $\frac{\text{FTA}}{\text{FGA}}$. How frequently a player gets to the free-throw line relative to their shot attempts.

Advanced Metrics

PER **Player Efficiency Rating.** A per-minute rating that summarizes a player's statistical accomplishments. League average is 15.0.

USG% **Usage Rate.** Estimates the percentage of team possessions a player uses while on the floor (via field goal attempts, free throw attempts, and turnovers).

BPM **Box Plus/Minus.** A box-score estimate of the points per 100 possessions a player contributes above a league-average player. Combines offensive and defensive components: $\mathrm{BPM} = \mathrm{OBPM} + \mathrm{DBPM}$.

OBPM **Offensive Box Plus/Minus.** The offensive component of BPM.

DBPM **Defensive Box Plus/Minus.** The defensive component of BPM. Our analysis found this to be the single strongest predictor of tournament success.

WS **Win Shares.** An estimate of the number of wins a player contributes to their team. Sum of OWS and DWS.

OWS **Offensive Win Shares.** Win Shares attributed to a player's offensive contributions.

DWS **Defensive Win Shares.** Win Shares attributed to a player's defensive contributions.

WS/40 **Win Shares Per 40 Minutes.** Win Shares normalized to a per-40-minute rate.

Percentage-Based Advanced Metrics

ORB% **Offensive Rebound Percentage.** Estimate of the percentage of available offensive rebounds a player grabs while on the floor.

DRB% **Defensive Rebound Percentage.** Estimate of the percentage of available defensive rebounds a player grabs while on the floor.

TRB% **Total Rebound Percentage.** Combined offensive and defensive rebound percentage.

AST% **Assist Percentage.** Estimate of the percentage of teammate field goals a player assisted while on the floor.

STL% **Steal Percentage.** Estimate of the percentage of opponent possessions that end with a steal by the player.

BLK% **Block Percentage.** Estimate of the percentage of opponent two-point field goal attempts blocked by the player.

TOV% **Turnover Percentage.** Estimate of turnovers committed per 100 plays.

Team-Level Metrics

SRS **Simple Rating System.** A team rating that accounts for average point differential and strength of schedule. An SRS of 0 is average; positive values indicate above-average teams.

SOS **Strength of Schedule.** A rating of the difficulty of a team's opponents. Derived from the SRS of all opponents faced.

Seed **Tournament Seed.** A team's seeding (1–16) in the NCAA tournament bracket, assigned by the selection committee based on regular-season performance. Lower seeds are stronger.

Statistical Methods

ROC-AUC **Area Under the Receiver Operating Characteristic Curve.** Measures how well a classifier distinguishes between classes across all probability thresholds. Ranges from 0.5 (random) to 1.0 (perfect).

Cohen's d **Effect Size.** Measures the standardized difference between two group means. Small: $|d| < 0.2$, Medium: $0.2 \leq |d| < 0.5$, Large: $|d| \geq 0.5$.

Bonferroni Correction
A method to control the familywise error rate when performing multiple simultaneous hypothesis tests, by dividing the significance threshold α by the number of tests.

Silhouette Score
A measure of how similar a data point is to its own cluster compared to other clusters. Ranges from -1 to $+1$; higher values indicate better-defined clusters.

BIC **Bayesian Information Criterion.** A model selection criterion that penalizes model complexity more strongly than AIC. Lower BIC indicates a better model.

Gini Importance
A feature importance measure for tree-based models. For each feature, it sums the total reduction in node impurity (Gini impurity) across all splits in all trees that use that feature.

Spearman's ρ
A non-parametric rank correlation coefficient measuring the monotonic relationship between two variables. Ranges from -1 to $+1$.

UMAP **Uniform Manifold Approximation and Projection.** A nonlinear dimensionality reduction technique that preserves both local and global data structure by constructing and optimizing a topological representation.

GMM **Gaussian Mixture Model.** A probabilistic model that represents the data as a weighted sum of Gaussian distributions, fitted via Expectation-Maximization. Provides soft cluster assignments.

DBSCAN **Density-Based Spatial Clustering of Applications with Noise.** A clustering algorithm that groups together points in high-density regions and marks points in low-density regions as noise. Does not require specifying the number of clusters.